전산화학 (2015년 1학기) 4차시험 정답

(총 **7문제, 110점**)

1. **(5점)**

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 6 & -1 & 2 \\ -1 & 3 & 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 1 & 4 \\ -1 & 3 & -1 & 4 \\ 2 & -6 & 2 & 2 \\ 3 & -7 & 3 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

등등.

2. **(10점)**

$$\begin{aligned} \mathbf{A}\langle -1,2,0\rangle \times \mathbf{B}\langle 2,2,0\rangle &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 2 & 0 \\ 2 & 2 & 0 \end{vmatrix}, \\ &= \mathbf{j}((2)\times(0)-(0)\times(2)) - \mathbf{j}((-1)\times(0)-(0)\times(2)) \\ &\quad +\mathbf{k}((-1)\times(2)-(2)\times(2)) \\ &= -6\mathbf{k}. \end{aligned}$$

0이고,

$$\begin{aligned} \mathbf{C}\langle 3,-1,0\rangle \times \mathbf{D}\langle 0,4,0\rangle &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 0 \\ 0 & 4 & 0 \end{vmatrix}, \\ &= \mathbf{i}((-1)\times(0)-(0)\times(4)) - \mathbf{j}((3)\times(0)-(0)\times(0)) \\ &\quad +\mathbf{k}((3)\times(4)-(-1)\times(0)) \\ &= 12\mathbf{k}. \end{aligned}$$

0이므로,

$$\boxed{(\mathbf{A}\times\mathbf{B})\cdot(\mathbf{C}\times\mathbf{D}) = (-6\mathbf{k})\cdot(12\mathbf{k}) = -72}$$

3. **(10점)**

$$
\begin{aligned}
\nabla^2 f(x,y,z) &= \left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial}{\partial z^2}\right)(\cos ax)(\cos by)(\cos cz) \\
&= (\cos by)(\cos cz)\frac{\partial^2}{\partial x^2}(\cos ax)+(\cos ax)(\cos cz)\frac{\partial^2}{\partial y^2}(\cos by)+(\cos ax)(\cos by)\frac{\partial^2}{\partial z^2}(\cos cz) \\
&= -a^2(\cos ax)(\cos by)(\cos cz)-b^2(\cos ax)(\cos by)(\cos cz)-c^2(\cos ax)(\cos by)(\cos cz) \\
&= -(a^2+b^2+c^2)(\cos ax)(\cos by)(\cos cz) \\
&= -(a^2+b^2+c^2)f(x,y,z)
\end{aligned}
$$

**그러므로 함수 $f(x,y,z)=(\cos ax)(\cos by)(\cos cz)$ 는 Laplacian 연산자의 고유함수이며, 고윳값은 $-(a^2+b^2+c^2)$ 이다.**

4. **(총 20점, 각 10점)**

(a)

$$
\begin{aligned}
\left[\frac{d}{dx},x^2\right]f(x) &= \frac{d}{dx}(x^2 f)-x^2\frac{d}{dx}f \\
&= (2xf+x^2f')-x^2f' = 2xf
\end{aligned}
$$

$$
\boxed{\left[\frac{d}{dx},x^2\right]=2x}
$$

(b)

$$
\begin{aligned}
\left[\frac{d^2}{dx^2}-x,\frac{d}{dx}+x^2\right]f(x) &= \left(\frac{d^2}{dx^2}-x\right)\left(\frac{d}{dx}+x^2\right)f-\left(\frac{d}{dx}+x^2\right)\left(\frac{d^2}{dx^2}-x\right)f \\
&= \left(\frac{d^2}{dx^2}-x\right)(f'+x^2f)-\left(\frac{d}{dx}+x^2\right)(f''-xf) \\
&= f'''+\frac{d}{dx}(2xf+x^2f')-xf'-x^3f-(f'''-f-xf'+x^2f''-x^3f) \\
&= f'''+2f+2xf'+2xf'+x^2f''-xf'-x^3f-f'''+f+xf'-x^2f''+x^3f \\
&= 3f+4xf' = \left(3+4x\frac{d}{dx}\right)f(x)
\end{aligned}
$$

$$
\boxed{\left[\frac{d^2}{dx^2}-x,\frac{d}{dx}+x^2\right]=3+4x\frac{d}{dx}}
$$

5. **(총 20점, 각 10점)**

(a)

$$\mathbf{I}_x^2+\mathbf{I}_y^2+\mathbf{I}_z^2 = \frac{1}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}+\frac{1}{\sqrt{2}}\begin{pmatrix}0&-i&0\\i&0&-i\\0&i&0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0&-i&0\\i&0&-i\\0&i&0\end{pmatrix}+$$
$$\begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix}\begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix}$$
$$= \frac{1}{2}\begin{pmatrix}1&0&1\\0&2&0\\1&0&1\end{pmatrix}+\frac{1}{2}\begin{pmatrix}1&0&-1\\0&2&0\\-1&0&1\end{pmatrix}+\begin{pmatrix}1&0&0\\0&0&0\\0&0&1\end{pmatrix}$$
$$= \begin{pmatrix}2&0&0\\0&2&0\\0&0&2\end{pmatrix} = 2\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$

(b)

$$\left[\mathbf{I}_x,\mathbf{I}_y\right] = \mathbf{I}_x\mathbf{I}_y-\mathbf{I}_y\mathbf{I}_x = \frac{1}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0&-i&0\\i&0&-i\\0&i&0\end{pmatrix}-\frac{1}{\sqrt{2}}\begin{pmatrix}0&-i&0\\i&0&-i\\0&i&0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}$$
$$= \frac{1}{2}\begin{pmatrix}i&0&-i\\0&0&0\\i&0&-i\end{pmatrix}-\frac{1}{2}\begin{pmatrix}-i&0&-i\\0&0&0\\i&0&i\end{pmatrix} = \begin{pmatrix}i&0&0\\0&0&0\\0&0&-i\end{pmatrix} = i\mathbf{I}_z \tag{1}$$

6. **(총 20점, 각 10점)**

(a)

$$x+2y-3z = \lambda x$$
$$2x+4y-6z = \lambda y$$
$$-x-2y+3z = \lambda z$$

위 연립방정식은 다음과 같이 재배열할 수 있다.

$$(1-\lambda)x+2y-3z = 0 \tag{2a}$$
$$2x+(4-\lambda)y-6z = 0 \tag{2b}$$
$$-x-2y+(3-\lambda)z = 0 \tag{2c}$$

위 연립방정식은 다음과 같이 행렬을 이용해 나타낼 수 있다.

$$\begin{pmatrix} 1-\lambda & 2 & -3 \\ 2 & 4-\lambda & -6 \\ -1 & -2 & 3-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

위 연립방정식이 비자명해를 갖기 위해서는

$$\begin{vmatrix} 1-\lambda & 2 & -3 \\ 2 & 4-\lambda & -6 \\ -1 & -2 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)\begin{vmatrix} 4-\lambda & -6 \\ -2 & 3-\lambda \end{vmatrix} - (2)\begin{vmatrix} 2 & -6 \\ -1 & 3-\lambda \end{vmatrix} + (-3)\begin{vmatrix} 2 & 4-\lambda \\ -1 & -2 \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)\big[(4-\lambda)(3-\lambda)-12\big] - (2)(-2\lambda) + (-3)(-\lambda) = 0$$

$$\Rightarrow \lambda^2(\lambda - 8) = 0 \tag{3}$$

(3) 식과 문제의 주어진 조건 $\lambda > 0$ 으로부터 $\lambda = 8$.

(b) (3) 식의 $\lambda = 8$ 을 (2a) - (2c) 식에 대입하여 정리하면 $y = 2x$, $z = -x$ 이다. 이를 주어진 조건 $x^2 + y^2 + z^2 = 1$ 에 대입하면 $6x^2 = 1 \Rightarrow x = \pm 1/\sqrt{6}$ 이다. 그러므로

$$x = \pm\frac{1}{\sqrt{6}}, \quad y = \pm\frac{2}{\sqrt{6}}, \quad z = \mp\frac{1}{\sqrt{6}}$$

7. **(25 점)**

(a) **(5 점)**

$$\hat{\mathbf{L}} = \frac{\hbar}{i}(\mathbf{r}\times\nabla) = \frac{\hbar}{i}\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix}$$

$$= \mathbf{i}\frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) - \mathbf{j}\frac{\hbar}{i}\left(x\frac{\partial}{\partial z} - z\frac{\partial}{\partial x}\right) + \mathbf{k}\frac{\hbar}{i}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

$$= \mathbf{i}\frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) + \mathbf{j}\frac{\hbar}{i}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right) + \mathbf{k}\frac{\hbar}{i}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

$$= \mathbf{i}\hat{L}_x + \mathbf{j}\hat{L}_y + \mathbf{k}\hat{L}_z, \tag{4}$$

그러므로,

$$\hat{L}_x = \frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \tag{5a}$$

$$\hat{L}_y = \frac{\hbar}{i}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \tag{5b}$$

$$\hat{L}_z = \frac{\hbar}{i}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right). \tag{5c}$$

(b) **(10점)** (5a) 식을 이용하면,

$$\begin{aligned}
\left[\hat{L}_x, \hat{y}\right] f(x,y,z) &= \hat{L}_x \hat{y} f(x,y,z) - \hat{y}\hat{L}_x f(x,y,z) \\
&= \frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) y f(x,y,z) - y\frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) f(x,y,z) \\
&= \frac{\hbar}{i}\left[\left(y^2\frac{\partial f}{\partial z} - zf - zy\frac{\partial f}{\partial y}\right) - \left(y^2\frac{\partial f}{\partial z} - yz\frac{\partial f}{\partial y}\right)\right] \\
&= -\frac{\hbar}{i} z f(x,y,z) = i\hbar f(x,y,z)
\end{aligned}$$

그러므로

$$\boxed{\left[\hat{L}_x, \hat{y}\right] = -\frac{\hbar}{i}\hat{z} = i\hbar\hat{z}}$$

(c) **(10점)** (5a) 식과 $\hat{z} = (\hbar/i)(\partial/\partial z)$을 이용하면,

$$\begin{aligned}
\left[\hat{L}_x, \hat{p}_z\right] f(x,y,z) &= \hat{L}_x \hat{p}_z f(x,y,z) - \hat{p}_z \hat{L}_x f(x,y,z) \\
&= \frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\frac{\hbar}{i}\frac{\partial}{\partial z} f(x,y,z) - \frac{\hbar}{i}\frac{\partial}{\partial z}\frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) f(x,y,z) \\
&= -\hbar^2\left[\left(y\frac{\partial^2 f}{\partial z^2} - z\frac{\partial^2 f}{\partial y \partial z}\right) - \left(y\frac{\partial^2 f}{\partial z^2} - \frac{\partial f}{\partial y} - z\frac{\partial^2 f}{\partial z \partial y}\right)\right] \\
&= -\hbar^2\frac{\partial f(x,y,z)}{\partial y} = -i\hbar\frac{\hbar}{i}\frac{\partial}{\partial y} f(x,y,z) = -i\hbar\hat{p}_y f(x,y,z)
\end{aligned}$$

그러므로

$$\boxed{\left[\hat{L}_x, \hat{p}_z\right] = -i\hbar\hat{p}_y}$$